Supplementary Material 3

Supplementary Material 3A: Stata code

```
*****************************************
//point estimate (sleep quality)
*****************************************
clear
input str50 study n m sd
"Bartholomei et al. 2018 (n=55)" 55 4.22 3.8
"Cupidi et al. 2012 (n=40)" 40 6.30 3.90
"Hicks et al. 2019 (n=18)" 18 4.94 2.15
"Liu et al. 2018 (n=36)" 36 5.19 2.99
"Olsson et al. 2016 (n=66)" 66 6.83 7.01
"Pal et al. 2004 (n=20)" 20 5.5 8.2
"Wade et al. 2020 (n=29)" 29 6.93 3.87
end

gen se = sd/sqrt(n)

meta set m se, studylabel(study) eslabel(Mean)
meta summarize
return list
local ES = r(theta)
meta forestplot, xline(`ES', lcolor(gs12) lpattern(dash)) noohom noosig xlabel(0(2)10) xline(0,
lcolor(gs7) lpattern(solid))

meta summarize, leaveoneout

*****************************************
//point estimate (sensitivity analysis: omit non-PSQI)
*****************************************
clear
input str50 study n m sd
"Cupidi et al. 2012 (n=40)" 40 6.30 3.90
"Hicks et al. 2019 (n=18)" 18 4.94 2.15
"Liu et al. 2018 (n=36)" 36 5.19 2.99
"Pal et al. 2004 (n=20)" 20 5.5 8.2
"Wade et al. 2020 (n=29)" 29 6.93 3.87
end

gen se = sd/sqrt(n)

meta set m se, studylabel(study) eslabel(Mean)
meta summarize
return list
```

```
local ES = r(theta)

meta forestplot, xline(`ES', lcolor(gs12) lpattern(dash)) noohom noosig xlabel(0(2)10) xline(0,
lcolor(gs7) lpattern(solid))

meta summarize, leaveoneout

*****************************************
//point estimate (sensitivity analysis: omit data transposed to mean[SD])
*****************************************
clear
input str50 study n m sd
"Bartholomei et al. 2018 (n=55)" 55 4.22 3.8
"Cupidi et al. 2012 (n=40)" 40 6.30 3.90
"Hicks et al. 2019 (n=18)" 18 4.94 2.15
"Liu et al. 2018 (n=36)" 36 5.19 2.99
"Pal et al. 2004 (n=20)" 20 5.5 8.2
"Wade et al. 2020 (n=29)" 29 6.93 3.87
end

gen se = sd/sqrt(n)

meta set m se, studylabel(study) eslabel(Mean)
meta summarize
return list
local ES = r(theta)

meta forestplot, xline(`ES', lcolor(gs12) lpattern(dash)) noohom noosig xlabel(0(2)10) xline(0,
lcolor(gs7) lpattern(solid))

meta summarize, leaveoneout

*****************************************
//Parkinsonism carers vs healthy control
*****************************************
clear
input str50 study n1 m1 s1 n2 m2 s2
"Cupidi et al. 2012" 40 6.30 3.90 150 4.3 3.1
"Olsson et al. 2016" 66 32.5 33.38 79 27.5 25.80
end

format %9.2f m1
format %9.2f s1
format %9.2f m2
format %9.2f s2
```

```
meta esize n1 m1 s1 n2 m2 s2, studylabel(study) eslabel(Hedges' g)
meta summarize, se(khartung)
return list
local ES = r(theta)

meta forestplot, xline(`ES', lcolor(gs12) lpattern(dash)) xlabel(-3(1)4)
nullrefline(favorsleft("Parkinsonism carers better ", al(baseline) height(45)) favorsright("
Parkinsonism carers worse", al(baseline) height(45))) columnopts(_data1,
supertitle(Parkinsonism carers)) columnopts(_data2, supertitle(Healthy control)) xline(0,
lcolor(gs7) lpattern(solid)) se(khartung)

****************************************
//Parkinsonism carers vs other carers
****************************************
clear
input str50 study n1 m1 s1 n2 m2 s2
"Liu et al. 2018 - FTLD carers" 18 5.19 2.99 131 4.4 3.70
"Liu et al. 2018 - AZ carers" 18 5.19 2.99 325 1.01 1.40
end

format %9.2f m1
format %9.2f s1
format %9.2f m2
format %9.2f s2

meta esize n1 m1 s1 n2 m2 s2, studylabel(study) eslabel(Hedges' g)
meta summarize, se(khartung)
return list
local ES = r(theta)

meta forestplot, xline(`ES', lcolor(gs12) lpattern(dash)) xlabel(-15(5)20)
nullrefline(favorsleft("Parkinsonism carers better ", al(baseline) height(45)) favorsright("
Parkinsonism carers worse", al(baseline) height(45))) columnopts(_data1,
supertitle(Parkinsonism carers)) columnopts(_data2, supertitle(Other carers)) xline(0,
lcolor(gs7) lpattern(solid)) se(khartung)
```

Supplementary Material 3B: point estimate (sleep quality; leave one out)

Omitted study	Mean	Lower 95%CI	Upper 95%CI
Bartholomei et al. 2018 (n=55)	5.871	5.097	6.645
Cupidi et al. 2012 (n=40)	5.484	4.550	6.418
Hicks et al. 2019 (n=18)	5.763	4.794	6.733
Liu et al. 2018 (n=36)	5.717	4.718	6.716
Olsson et al. 2016 (n=66)	5.435	4.580	6.289
Pal et al. 2004 (n=20)	5.617	4.747	6.488
Wade et al. 2020 (n=29)	5.365	4.563	6.166

Supplementary Material 3C: point estimate (sensitivity analysis: omit non-PSQI)

[INSERT SUPPLEMENTARY MATERIAL 3C FIGURE]

Supplementary Material 3D: point estimate (sensitivity analysis: omit non-PSQI; leave one out)

Omitted study	Mean	Lower 95%CI	Upper 95%CI
Cupidi et al. 2012 (n=40)	5.565	4.571	6.559
Hicks et al. 2019 (n=18)	6.000	5.064	6.937
Liu et al. 2018 (n=36)	5.931	4.870	6.993
Pal et al. 2004 (n=20)	5.739	4.861	6.618
Wade et al. 2020 (n=29)	5.402	4.708	6.097

Supplementary Material 3E: point estimate (sensitivity analysis: omit data transposed to mean[SD])

[INSERT SUPPLEMENTARY MATERIAL 3E FIGURE]

Supplementary Material 3F: point estimate (sensitivity analysis: omit data transposed to mean[SD]; leave one out)

Omitted study	Mean	Lower 95%CI	Upper 95%CI
Bartholomei et al. 2018 (n=55)	5.720	4.899	6.542
Cupidi et al. 2012 (n=40)	5.244	4.303	6.185
Hicks et al. 2019 (n=18)	5.577	4.517	6.636
Liu et al. 2018 (n=36)	5.519	4.430	6.607
Pal et al. 2004 (n=20)	5.437	4.528	6.346
Wade et al. 2020 (n=29)	5.129	4.378	5.879